

Contents

1 Finite-depth Lyapunov-growth estimation in GNNs	1
1.1 Executive conclusions	1
1.2 Estimators: what each algorithm estimates	1
1.3 Controlled GNN/data study	2
1.4 Verification before measurement	2
1.5 Experience 1: random probes converge far too slowly here	3
1.5.1 ChebNet replication	4
1.6 Experience 2: power iteration succeeds on the same finite products	5
1.7 Experience 3: rates, graph families and training state	5
1.8 Recommended default protocol	5
1.9 Limits and next experiments	7
1.10 Evidence and reproducibility	7
1.11 References	8

1 Finite-depth Lyapunov-growth estimation in GNNs

Status: reproducible pilot study, 2026-09-24.

Question: how should we estimate the top depth-wise Jacobian growth rate of a trained GNN, and how dependable are random tangent probes across GNNs and graph/data conditions?

This report concerns the finite network quantity

$$\lambda_K^{\text{top}}(G, x; \Phi) = \frac{1}{K} \log \sigma_1(B_K), \quad B_K = D_{h_0}(F_{K-1,G} \circ \dots \circ F_{0,G})(h_0),$$

where h_0 is the output of the encoder and h_K the output of the last hidden propagation layer. It does **not** include encoder or decoder derivatives and is not automatically an infinite-depth Lyapunov exponent. All results use the complete layer derivative at the forward state, including activation and state-conditioned sheaf restriction-map derivatives.

Non-negotiable implementation detail: no tangent vector is mean-centered or projected between layers. Pierre’s P -projected procedure measures a different product, not B_K . Scalar norm renormalization is used only to prevent numerical overflow/underflow.

1.1 Executive conclusions

1. **A maximum over independent JVP probes is correct but was not useful as a point estimate here.** It is a rigorous lower bound for every finite M and converges almost surely as $M \rightarrow \infty$. Yet even **1,024** probes did not meet the predeclared 0.01-per-layer error target in any of the 432 trained calibration streams: pooled median gaps remained 0.395 for GCN and 0.109 for the deep sheaf baseline.
2. **Matrix-free power iteration on $B_K^* B_K$ is the right default for the top finite-depth rate when VJPs are available.** With three starts and 160 iterations, all 36 tiny trained calibration cells met residual $\leq 10^{-7}$ and agreed with reverse-mode dense-SVD references to mean absolute rate error 3.9×10^{-17} (maximum 2.2×10^{-16}).
3. **Probe count is architecture/spectrum dependent, not a universal GNN hyperparameter.** In this study, random probes were looser for the depth-3 GCN and depth-3 ChebNet than for the depth-10 sheaf baseline despite comparable tangent dimensions; graph family had a smaller effect than architecture. Training did not make random probes reliably tighter in the initialization ablation.
4. **Do not infer oversquashing or task utility from this top rate alone.** The GCN and sheaf baseline have positive top hidden-state rates here while their test accuracy differs sharply under a heterophilic distribution shift. Top global sensitivity does not identify source-to-target communication or spatial delocalization.

The findings are strong evidence for estimator choice on these controlled models, not a claim that all GNNs or real-world datasets behave identically. Broader datasets, architectures and hyperparameter budgets are the next research phase.

1.2 Estimators: what each algorithm estimates

Algorithm	Quantity	What is rigorous	Practical verdict
Dense Jacobian + SVD	Exact λ_K^{top}	Exact finite- K reference	Use only at tiny tangent dimension
M full-space JVP probes	$\max_{m \leq M} K^{-1} \log \ B_K q_m\ $	Finite- M lower bound; a.s. convergence as $M \rightarrow \infty$	Use as a lower-bound/spectral-shape diagnostic, not the default point estimate
JVP/VJP power on $B_K^* B_K$	Approximation to λ_K^{top}	Standard power convergence conditional on start overlap and spectral separation; final residual is directly checkable	Default top-rate estimator
Lanczos/Krylov on $B_K^* B_K$	Approximation to the same top singular value	Matrix-free extremal-eigenvalue method	Escalate to it for poor power residuals or clustered top spectrum
QR/Benettin frame	Long-depth Lyapunov spectrum under an extendable cocycle	Standard numerical asymptotic-spectrum method	Not interchangeable with finite- K singular values
Layerwise singular norms/product bounds	An upper bound or a different diagnostic	Not $\sigma_1(B_K)$ in general	Never report as this exponent
Finite nonlinear perturbations	Secant growth	Depends on perturbation scale and activation region	Optional autodiff audit only

The key random-probe obstruction is visible in the rank-one case. If $B = \sigma uv^\top$ and q is uniform on S^{D-1} , then $\|Bq\|/\sigma = |v^\top q|$ and $(v^\top q)^2 \sim \text{Beta}(1/2, (D-1)/2)$. Typical overlap is $O(D^{-1/2})$. Thus a narrow leading direction can require very many parallel probes even though a power step aligns it immediately. A broad near-top singular subspace produces the opposite behavior. This is why a probe budget cannot be fixed from depth alone.

1.3 Controlled GNN/data study

Exact equations and scope are in [notes/experiment-baselines.md](#). We used:

- **GCN:** a depth-3 Kipf–Welling stack, $h_{\ell+1} = \text{ReLU}(\tilde{A}h_\ell W_\ell)$, with self-looped symmetric normalized adjacency.
- **ChebNet:** depth 3 and Chebyshev order 3, with the self-loop-free symmetric normalized Laplacian and $\lambda_{\max} = 2$ scaling; it supplies a polynomial-filter contrast to one-hop GCN propagation.
- **Deep feature-conditioned diagonal-sheaf baseline:** depth 10, endpoint-dependent diagonal restriction maps, an assembled sheaf-Laplacian action and a residual ReLU update. It is inspired by Bodnar et al.’s neural sheaf diffusion, but is **not** claimed to reproduce every detail of that paper. Its implemented normalization is one-sided row normalization, not symmetric $D^{-1/2}L_{\mathcal{F}}D^{-1/2}$.
- **Task/data:** a balanced contextual two-community SBM for training/validation; held-out homophilic SBM, heterophilic SBM, and barbell graph conditions for testing. Two seeds, validation-only selection from widths $\{8, 12\}$ and learning rates $\{0.003, 0.01\}$, 120 Adam epochs.
- **Calibration:** three held-out trajectories per condition per seed. Reverse-mode dense autodiff Jacobian/SVD is the tiny-case reference. Full-space float64 unit probes use nested streams; no P projection is present.

All three models learned the in-distribution task comparably (mean held-out homophilic accuracy: GCN 0.865; ChebNet 0.861; sheaf 0.878). Under the heterophilic shift, ChebNet scored 0.524, the sheaf baseline 0.469, and GCN 0.312; all were high on the barbell condition (0.964, 0.958 and 0.944 respectively). These accuracy values validate that the models were not merely untrained random maps, but they are not a broad architecture benchmark.

1.4 Verification before measurement

The implementation has Slurm-verified unit tests at clean commit `c909dd60d39d51e90ad15413af8470b0da4735fd` (job 22347; 8 passed; evidence in `RESEARCH_LOG.md`):

- explicit linear product equals the JVP-built product;
- zero product returns $-\infty$;
- an actual regular-graph GCN constant mode survives (detecting an accidental P projection);

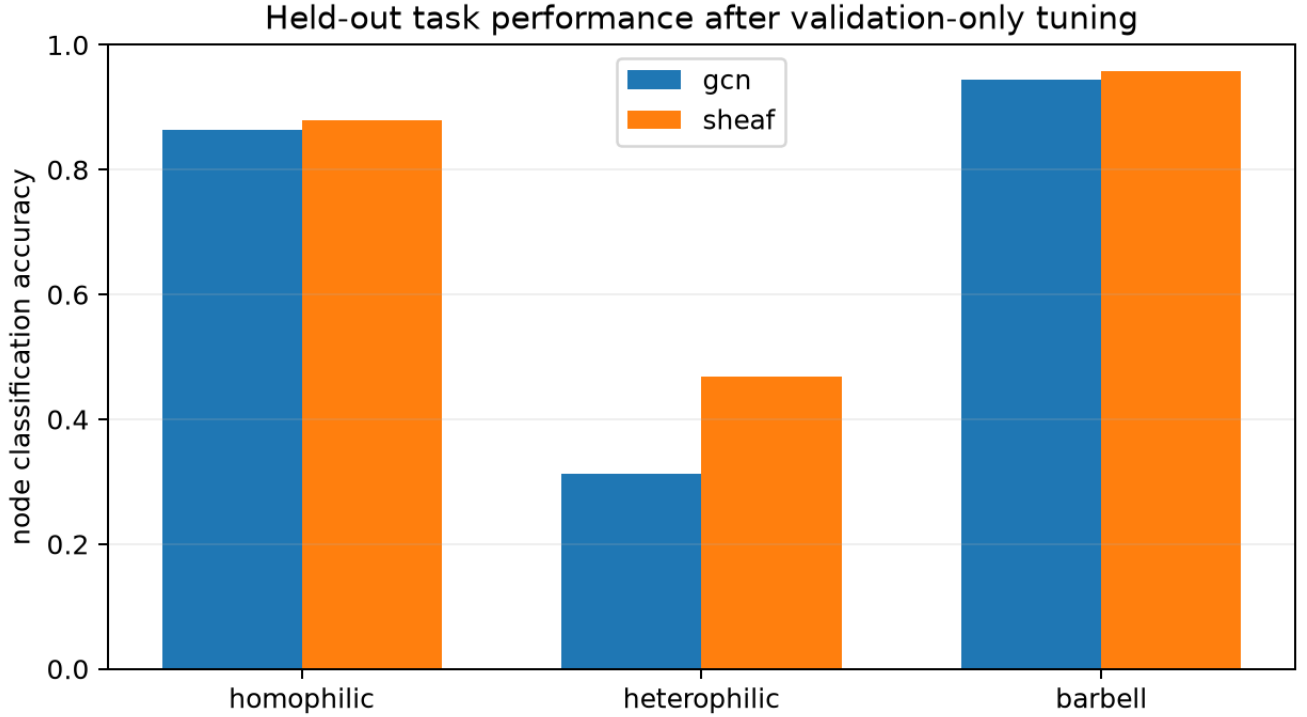


Figure 1: Held-out task accuracy

- non-unit supplied probes are rejected;
- power agrees with dense SVD on a known linear product;
- nonlinear GCN and sheaf JVP products agree with an **independent reverse-mode dense autodiff Jacobian**.
The declared ReLU convention is PyTorch’s derivative zero at a kink.

For every small experimental cell, the maximum entrywise discrepancy between JVP-built and reverse-mode product was at most 2.2×10^{-16} for GCN and 2.1×10^{-14} for sheaf. This validates autodiff implementations, but not a hypothetical shared framework bug; it is why both a structural test suite and two AD modes are retained.

1.5 Experience 1: random probes converge far too slowly here

The v2 study used 12 independent nested streams through each trained cell, with $M = 1, 2, 4, \dots, 1024$. Shading is the 5th–95th percentile over all selected seeds, trajectories and streams within a graph family.

At $M = 1$, pooled median gap was 0.847 (GCN) and 0.231 (sheaf). At $M = 128$, it was 0.449 and 0.127. At $M = 1024$, it was 0.395 and 0.109. The target was 0.01 per layer, so the relevant result is not merely that convergence is slow: **coverage at the target was 0/72 for every model–dataset family at $M = 1024$.**

Model	Dataset family	Median gap at $M = 1024$	95th percentile gap	Target coverage
GCN	homophilic	0.376	0.452	0/72
GCN	heterophilic	0.379	0.469	0/72
GCN	barbell	0.417	0.473	0/72
Sheaf	homophilic	0.107	0.117	0/72
Sheaf	heterophilic	0.109	0.119	0/72
Sheaf	barbell	0.112	0.124	0/72

The per-family plots make clear that dataset family changes the curve somewhat, but the architecture-level difference is larger in this pilot:

Random-probe lower-bound convergence by architecture and graph family

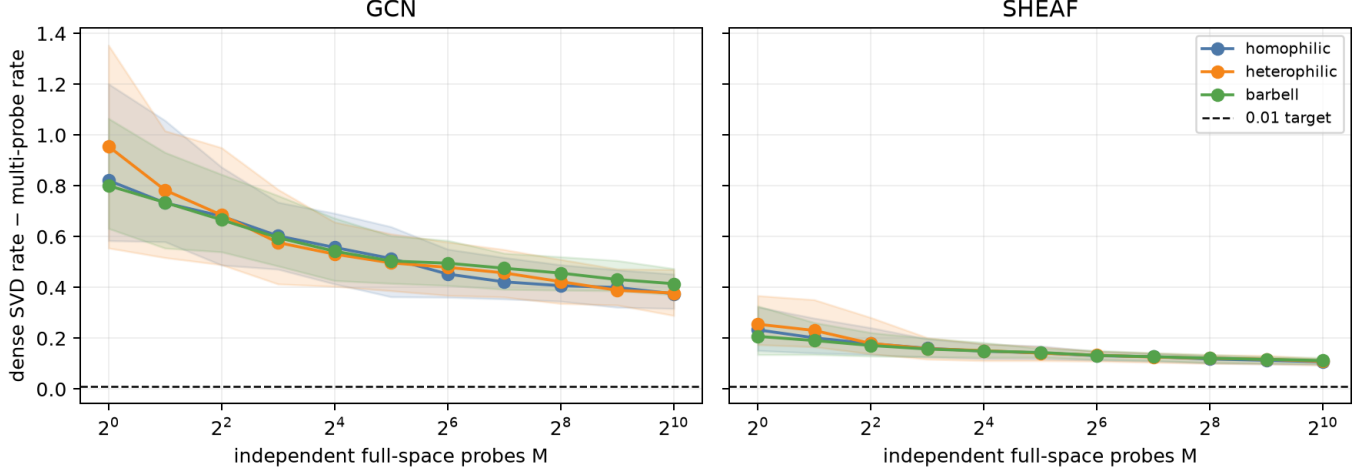
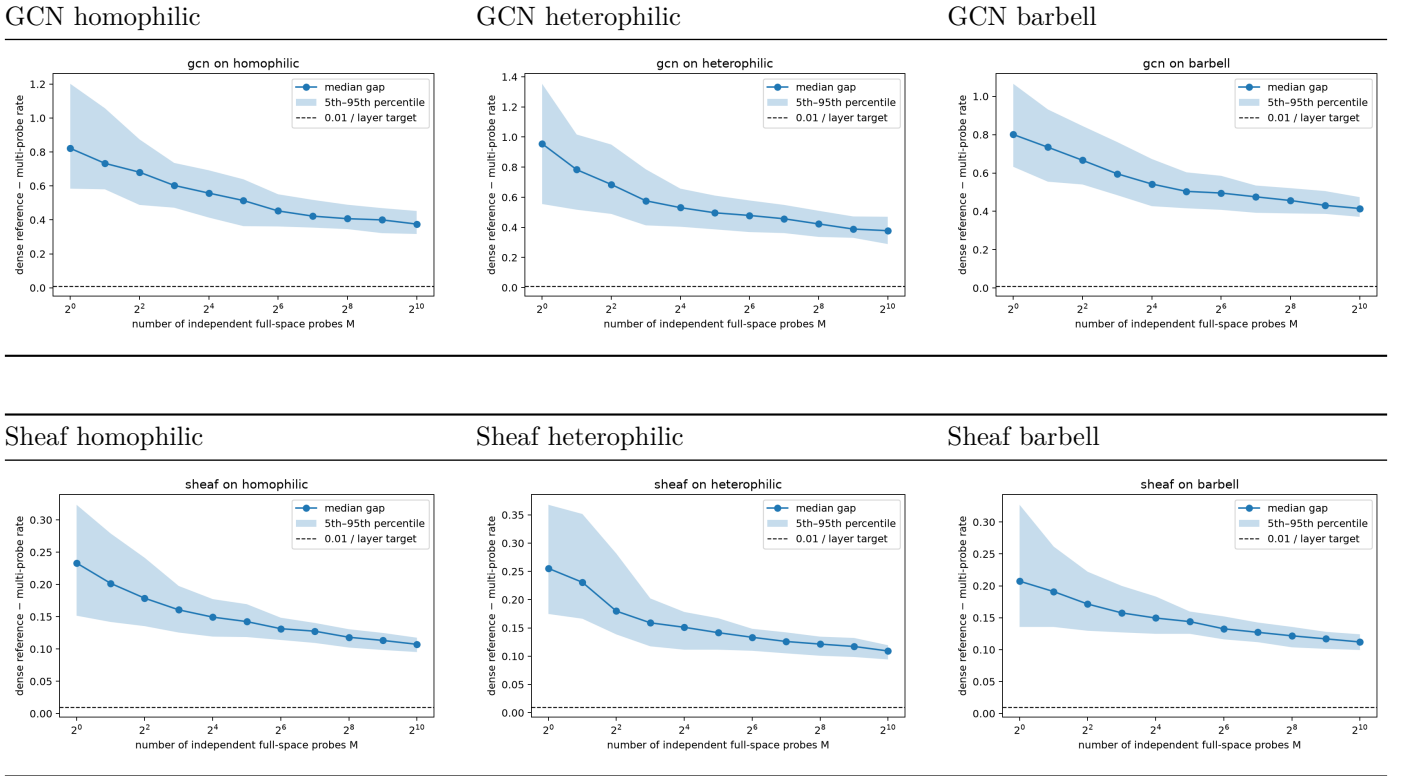


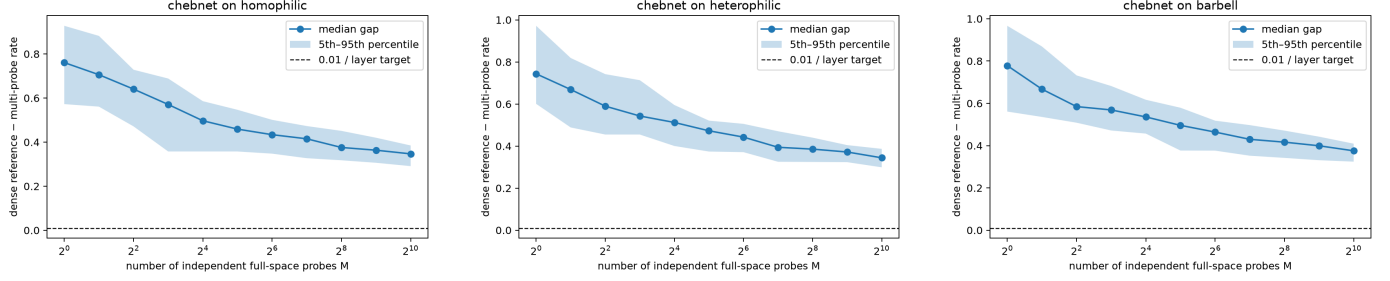
Figure 2: Probe convergence summary



Interpretation: these data falsify the initially tempting default “use 128 random probes and call the maximum the exponent.” They do *not* say that a random-probe lower bound is wrong; it is doing exactly what it promises. The high gap suggests that the dominant finite-product direction is fairly concentrated relative to random full-space directions, especially for these trained GCNs and ChebNets.

1.5.1 ChebNet replication

A separately tuned depth-3/order-3 ChebNet calibration (two seeds, three trajectories per graph family, 12 streams through 1024 probes) replicates the random-probe conclusion: median $M = 1024$ gaps were 0.347 (homophilic), 0.346 (heterophilic), and 0.376 (barbell), with 0/72 streams at the 0.01 target in every family. All 18 corresponding three-start/160-iteration power cells met the residual criterion. Its curve lies between the GCN and sheaf cases, showing that the result is neither specific to a one-hop filter nor explained only by depth.



1.6 Experience 2: power iteration succeeds on the same finite products

For the same 36 trained tiny cells, three random-start power runs of $160 B_K^* B_K$ iterations were performed. The best returned rate was accepted only when its explicit relative residual was $\leq 10^{-7}$. All 36 cells met the condition. Compared with the reverse-mode dense SVD rate, the mean absolute error was 3.9×10^{-17} and maximum 2.2×10^{-16} ; these are tiny-reference agreement results, not an assertion that 160 iterations will suffice at every scale.

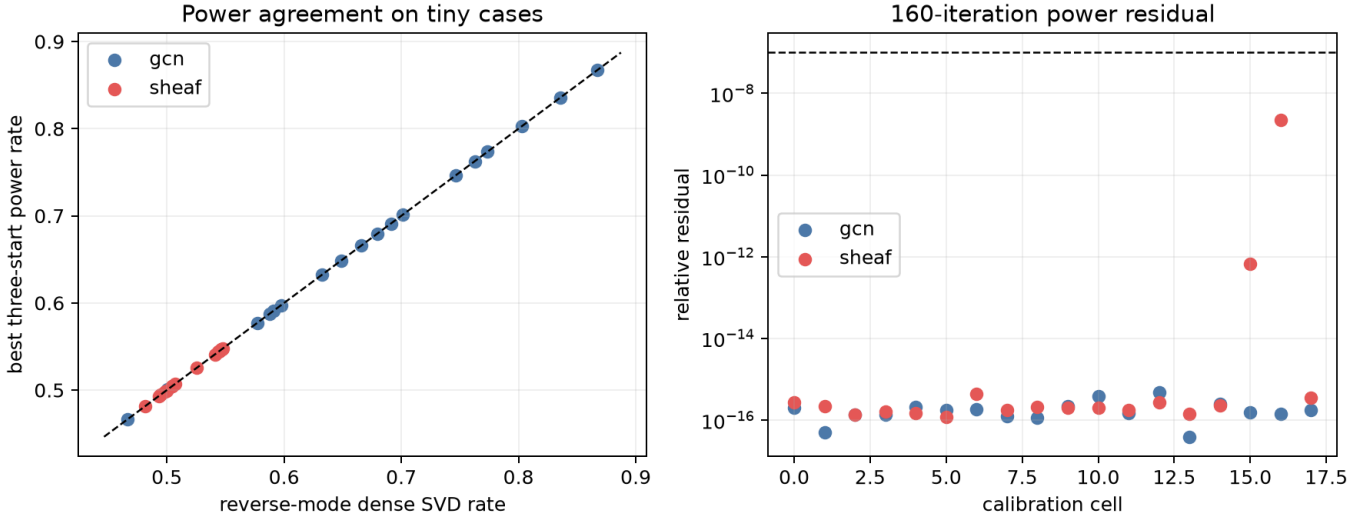


Figure 3: Power versus exact SVD

This contrast is expected: power iteration actively rotates a generic nonzero-overlap start toward the dominant right singular direction, whereas independent raw probes only wait to sample a close one. The final residual is essential. In v1, a single 80-iteration start failed the 10^{-7} residual criterion in two deep-sheaf/barbell cells; v2's three-start/160-iteration protocol resolved all tiny cells. A low residual validates convergence of the chosen Krylov iterate, while agreement of multiple starts helps reveal a poor start or numerical pathology.

1.7 Experience 3: rates, graph families and training state

The exact top rate is positive in this pilot. That means selected hidden perturbations can grow across depth; it does not by itself predict prediction accuracy or distant-node sensitivity.

A separate trained-versus-untrained ablation used two seeds, three trajectories per family, eight streams and up to 512 probes. Training did not make raw probe estimates tight. At $M = 512$, the GCN median gaps ranged from 0.418–0.430 trained and 0.396–0.418 untrained; sheaf gaps ranged from 0.111–0.116 trained and 0.111–0.114 untrained. This is useful negative evidence: random-probe behavior is not safely inferred from whether a model has learned the task.

1.8 Recommended default protocol

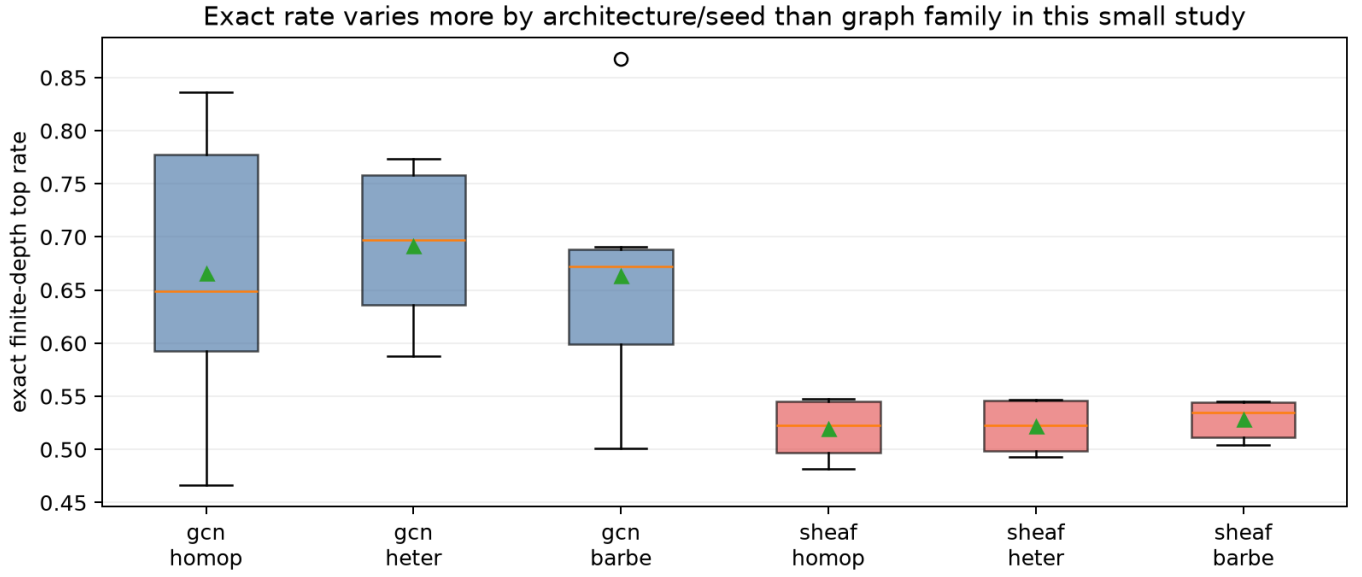


Figure 4: Exact-rate distribution

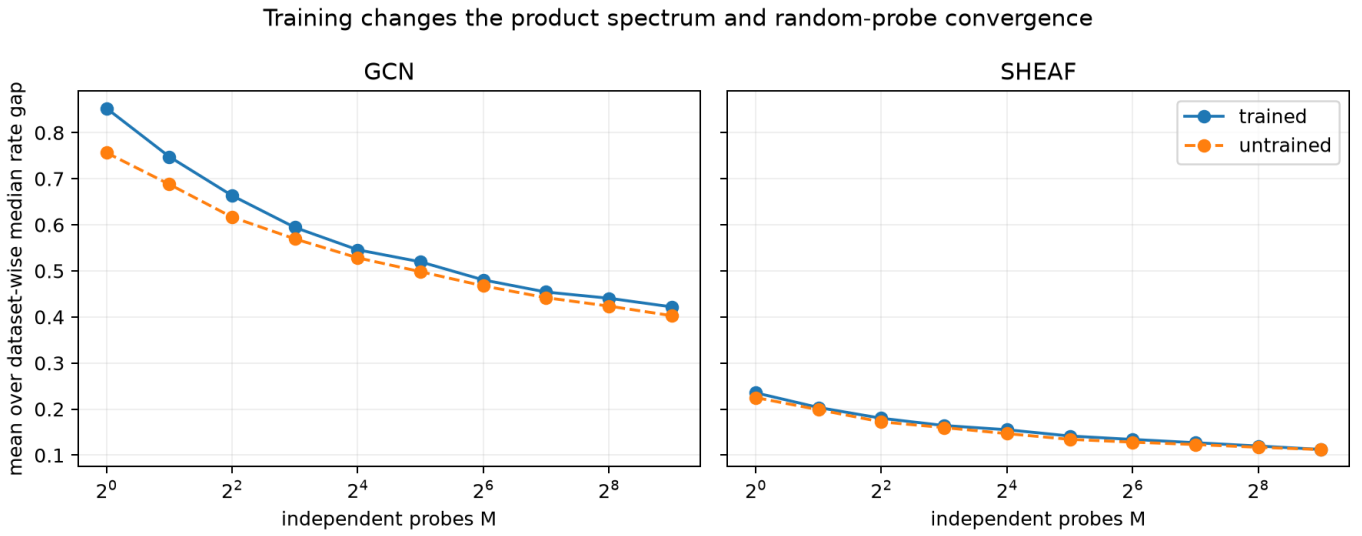


Figure 5: Training-state probe comparison

Situation	Recommendation	Required reporting
Tiny calibration, e.g. $D \lesssim 200$	Reverse-mode dense Jacobian/SVD plus JVP-product agreement	Both AD routes, max product discrepancy, exact singular value/rate
Routine finite- K top rate with JVP and VJP	Three-start power iteration on $B_K^* B_K$; 160 iterations initially; accept only residual $\leq 10^{-7}$	each start’s rate/residual, accepted start, JVP/VJP sweep count, precision
Power fails residual/start agreement	Run more iterations; then matrix-free Lanczos/Krylov; otherwise label result unresolved	residual history and stop reason; never silently emit a number
JVP-only setting	Report nested multi-probe maxima explicitly as lower bounds; calibrate M against a trusted reference by architecture/dimension	M , stream count/seeds, gap coverage curve; no unqualified “exponent” claim
Long, tied/stationary depth process and spectrum needed	QR/Benettin frame, with an explicitly justified stationary cocycle evaluation mode; fixed graph and stochastic masks; full derivative through adaptive geometry/normalization; no P	transient/renormalization schedule and convergence history
Any GNN		state boundary, hidden width/depth, dtype, graph/example IDs, full commit and lock hashes

The 160/three-start choice is an evidence-supported **initial default**, not a permanent magic number. At larger dimensions, the residual threshold and repeat-start checks—not the iteration count—should decide acceptance. Lanczos is the next algorithmic upgrade because it uses the same JVP/VJP primitives but can improve convergence for an unfavorable spectrum.

1.9 Limits and next experiments

- The graphs are deliberately small because exact SVD is central to calibration. We have not established a power-iteration residual threshold on large real-world GNNs.
- The sheaf architecture is an explicit diagnostic baseline, not a faithful reproduction claim for every Neural Sheaf Diffusion variant. The state-dependent- μ GNN remains underspecified in `notes/state-dependent-mu-diffusion-GNN.md`.
- The study is node classification on synthetic graph distributions; it does not yet cover citation/coauthor/molecular benchmarks, graph-level tasks, dropout/DropEdge or heterogeneous feature sizes.
- Global $\sigma_1(B_K)$ cannot diagnose particular source–target blocks of the Jacobian, delocalization, oversmoothing or oversquashing. Those need separate controlled observables.
- The next scale-up should retain this exact-SVD calibration suite, then test power/Lanczos residual behavior and rate distributions on tuned GCN, ChebNet, a specified sheaf implementation, and the specified adaptive- μ model on real benchmarks. Hyperparameters must be selected from validation data only.

1.10 Evidence and reproducibility

- **v1** (up to 128 probes): commit `e33f696c12a84d914ca008d99405a2cccd70d89f`, Slurm 22348; raw `results/2026-09-24-probe-calibration-v1/`; evidence `evidence/2026-09-24-probe-calibration-v1/`.
- **v2** (up to 1024 probes; 3×160 power): commit `12bed9354b76c11fea3bdf4cd1f4244b68a8e6d7`, Slurm 22351; raw `results/2026-09-24-probe-calibration-v2/`; evidence `evidence/2026-09-24-probe-calibration-v2/`.
- **Initialization ablation**: commit `151934db9d8c3a67eef1d1be258715bb68d773be`, Slurm 22355; raw `results/2026-09-24-probe-calibration-initialization/`; evidence `evidence/2026-09-24-probe-calibration-initialization/`.
- **ChebNet replication**: commit `1224c6b5d1d7e557ac42938183232bcfa4c32e3d`, Slurm 22359; raw `results/2026-09-24-probe-calibration-chebnet/`; evidence `evidence/2026-09-24-probe-calibration-chebnet/`.

All runs used Python 3.12.3, PyTorch 2.14.0+cu130 in float64; the lockfile SHA-256 is `f1fc70f6db931bcdffb94d353d7ff9e70ed1bc`. Curation jobs were 22352/22353/22356. Full manifests state seeds, selected hyperparameters, tensor dimensions, training metrics, power residuals and environment. `RESEARCH_LOG.md` holds command/log paths and prior validation evidence.

1.11 References

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- Defferrard, Bresson & Vandergheynst, *Convolutional Neural Networks on Graphs with Fast Localized Spectral Filtering*, arXiv:1606.09375. Local source: [papers/1606.09375/](#).
- Bodnar et al., *Neural Sheaf Diffusion*, arXiv:2202.04579. Local primary source: [papers/2202.04579/](#).
- Oono & Suzuki, *Graph Neural Networks Exponentially Lose Expressive Power for Node Classification*, arXiv:1905.10947. Local source: [papers/1905.10947/](#). Its contraction result concerns distance to an invariant/smoothed subspace under assumptions; it is not a license to insert P into our estimator.
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- Benettin et al., *Lyapunov Characteristic Exponents for Smooth Dynamical Systems*, Meccanica 15, 1980. QR/orthonormal-frame context for asymptotic spectra.